

Cryptocurrency as money: A trading strategy solution

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Abstract

This paper is motivated by a hypothesis that the long term value of a cryptocurrency is determined by its future use as money. For a cryptocurrency to be used as a medium of payment, it has to fulfill three independent functions: medium of exchange, a unit of account, and store of value. Currently, cryptocurrencies are held for investment purposes rather than being used for transactions and thus as a medium of exchange. For cryptocurrency to become widely adopted as a means of payment, it first needs to go through a very volatile period because speculative traders see long-run future value in the cryptocurrency. In order to soften transition from speculative asset to medium of payment a trading strategy is proposed, which provides liquidity and reduces volatility. Similar to pairs trading strategy, the proposed solution is based on cointegration and performed in three steps. The main difference is that proposed solution does not include shorting, but holding cryptocurrencies, thus increasing the total available cash and adding to the equilibrium price. Results from an ongoing experiment suggest that the proposed trading strategy is appealing for about 40% of cryptocurrency investors, as the struggle against volatility problem is accompanied by significant financial gains.

Keywords: bitcoin, cryptocurrency, volatility, trading strategy

JEL Classification: G11, G14

1 Introduction

This paper proposes a trading strategy as a solution to the inherent volatility problem of cryptocurrencies; resolving it might pave the way to potential wide-spread adoption. The proposed method is a “disarmingly simple” trading strategy, based on the state-of-the-art findings from financial and theoretical literature. We lean on an abstract model of a single cryptocurrency market with two types of traders: a noise trader and a chartist, which are usually included in the modeling of artificial financial markets. Our understanding is that a long term growth of cryptocurrency price is determined by the inflow of cash into the market and that the long term value of a cryptocurrency is determined by its future use as money.

For a cryptocurrency to be used as a medium of payment, it has to fulfill three independent functions: medium of exchange, a unit of account, and store of value. A good store of value means that they can be reliably saved, stored, and retrieved while retaining its usefulness over time. Cryptocurrencies usually perform well as a medium of exchange with low transaction fees and peer-to-peer, global and government free design. However, good money should also provide stable prices to best perform its role as a unit of account, which enables well-informed economic decisions by households and firms: saving, borrowing, consumption, and investment are approached with the confidence that the value of one currency unit will be stable over time. Are cryptocurrencies used as money? Baur et al. (2018) focusing on Bitcoin, report that about 30% of coins are held by investors and the minority of users, both in number and Bitcoin balances, appear to use Bitcoin as a medium of exchange. This suggests that at present cryptocurrencies are held for investment purposes rather than being used for transactions and thus as a medium of exchange.

There might be a paradox, that for cryptocurrency to become widely adopted as a means of payment, it first needs to go through a very volatile period because speculative traders see long-run future value in the cryptocurrency. They take speculative positions, pushing up fees and prices, and crowding out people and institutions, who would want to use it as a medium of payment. Eventually, as the price stabilizes, gains from speculation are reduced and thus leading to lower fees paid. Stabilization allows for the beginning of use as a means of payment. Although no cryptocurrency is currently being widely accepted as a payment medium, speculation on future values may nevertheless be rational. In a simplistic view of a cryptocurrency market, the equilibrium price is equal to the ratio of total available cash to overall available cryptocurrency in the market. If these quantities were constant, there would be a mean reversion of prices. If new traders bring in more cash than new mined cryptocurrency, the price will increase, and the rest is just trading noise. For illustration purposes, traders can be segmented into two groups:

1. **Noise traders.** They place orders on the exchange randomly without regard to price or any other information they may possess about the long-term value of the traded asset. There are also several theoretical papers suggesting that noise trading increases the volatility of asset returns (e.g. Kyle, 1985; Black 1986; Delong et al., 1990; Campbell & Kyle, 1993).
2. **Chartists.** They represent speculators, aimed to gain by placing orders in the cryptocurrency market. They speculate that if prices are moving in one direction, they will continue to do so (e.g. Vigfusson, 1997, Lux & Marchesi, 1998; Westerhoff & Reitz, 2003).

Efficient Market Hypothesis states that asset prices fully reflect all past and current publicly available information (Fama, 1970). However, in practice, markets have shown not to be perfectly efficient. The EMH consist of three levels of efficiency: Strong, Semi-strong, and Weak. The Strong form assumes that all private information is fully integrated into the prices. The Semi-strong form implies that all of the public information is incorporated in market prices, while the weak form assumes that only past information is reflected in market prices and future prices follow a random walk (Fama, 1970). Kyröläinen (2008) finds that day trading by individual investors is positively associated with volatility even when general trading activity is controlled for. The volatility effect of day trading is consistent across sub-periods. On the other hand, day trading by institutional investors seems to be only weakly related to volatility. These findings suggest that less informed trades by individuals have a more significant impact on volatility than informed institutional trades, supporting the theoretical models in which noise trading affects prices.

Another feature of cryptocurrency prices are the extreme fat tails of the return distribution (e.g., Zargar & Kumar, 2019), which are attributed to the distribution of traders' wealth, that impact the size of the orders, and the "herding" effect of Chartists when a price trend is established (e.g. Wagner, 2003). Evidence of herding in the cryptocurrency market shows that cryptocurrency traders are likely to follow the decisions of other investors, especially as uncertainty increases (Bouri et al., 2018). Such investor behavior harms market efficiency. In addition to the effect of day trading on cryptocurrency volatility, empirical evidence also supports the conclusion that blockchain congestion is a factor in explaining volatility (Zimmerman, 2018). Cocco et al. (2017) experimented with an agent-based model of artificial exchange. Their computational experiments produced price series for which they were unable to reject the hypothesis that they follow a random walk. Their results depended on the presence of at least one kind of trader type - Noise traders, which was previously argued by Raberto et al. (2001). A cryptocurrency market is thus only weakly efficient.

In a simple model, a constancy of cryptocurrency amount and fiat money would imply that the rates would converge to the equilibrium price - to the ratio of total available cash to entire available cryptocurrency. Corbet & Katsiampa (2018) examined whether

Bitcoin price returns exhibit reverting patterns for minutely, hourly, daily, and weekly gains. They found evidence of asymmetric reverting patterns in the Bitcoin price returns. Moreover, they found stronger reverting behavior of negative price returns in terms of both reverting speed and magnitude compared to positive returns, irrespective of the period considered. Extending to multiple cryptocurrencies, Sovbetov (2018) has shown that the prices of cryptocurrencies are cointegrated and do not separate in the long run; converge to long-run equilibrium. Unexpected news, changes in supply, and demand are not efficiently incorporated in cryptocurrency prices, making price temporarily deviating from its equilibrium price. During deviation, arbitrage opportunities exist for market participants. Different strategists apply different tactics to detect and exploit these market anomalies.

The most common strategy dealing with cointegrated prices is pairs trading strategy. Gatev et al. (1999) introduced the distance method, where pairs are formed by underlying co-movement of prices by a so-called distance.. An asset (e.g. cryptocurrency) is paired with a matching asset when their distance is relatively small. A trade is executed when the prices deviate at least two standard deviations from its mean. By using this simple standard deviation strategy, Gatev et al. (1999) achieve profits net exceed transaction costs, thus creating a net profitable strategy. Later, they followed up on their earlier work using a more recent dataset and confirmed net profitability of this method (Gatev et al., 2006). The basis of the pairs trading is cointegration, meaning that the prices of two instruments will revert to mean. The idea behind pairs trading is “disarmingly simple” (Gatev et al., 2006). Go long the first component while going short the second one, when their spread widens, and if the history repeats itself, the prices will converge - yielding profits. Psaradellis et al. (2019) investigate pairs trading strategy based on the assumption of cointegrated prices in an informationally efficient market. They employ technical analysis to predict the prices movements of formatted spreads. Their findings reveal that technical trading has predictive power for numerous trading pairs.

2 Methods

The proposed solution takes into account the main conclusions of the literature to propel cryptocurrencies to wide-spread adoption and therefore:

- Provide liquidity
- Reduce volatility
- Increase market efficiency

Similar to pairs trading strategy, the proposed solution is based on cointegration and performed in three steps. The main difference is that we do not suggest shorting, but holding cryptocurrencies, thus increasing the total available cash and adding to the equilibrium price. The first step requires buying two (competing) cryptocurrencies (e.g.,

Bitcoin and Litecoin). In the second step, the trader places a series of limit orders above and below the market price within a selected range, which renders the strategy from relative-value arbitrage to liquidity providing strategy (see Figure 1 below for visual clarification of the proposed strategy). Noise traders and chartists will ensure that the price will move one way or the other, depending on the randomly occurring trend from the random traders and magnified by the chartists. When the price reversion kicks in the trader will be able to buy back previously sold cryptocurrency and buy more than he had before. Therefore each random move from the equilibrium price will provide gain on when the price reverses and lessen the effect of chartists and noise traders in subsequent cycles. This strategy offers liquidity and improves overall market efficiency, resulting in a financial benefits to the users until the market stabilizes.

To test the proposed solution, we have developed an online automated trading application and tracked the performance of the users. Users had the option to adjust the necessary parameters (number of steps - sell and buy orders, and trading range) and select the desired trading pair or pairs. Users were presented a choice of default settings – 30% trading range and seven steps regardless of the chosen trading pair. A backtest of the trading strategy is shown in Figure 1. The strategy sets out by buying two cryptocurrencies in equal amounts (in cash value) and placing orders with linearly increasing quantity on both sides (buy and sell). Figure 1 is composed of 4 charts: LTC/BTC illustrates buy and sell orders, together with the “automated split” cycle reset - a point at which application equally splits the funds again with a profit and resets the orders. Growth chart represents the total value expresses in each of the currencies, Asset allocation shows the percent of each cryptocurrency held during a one year period, while the Volume shows total traded volume assuming an initial investment of 1 Bitcoin.

Users had to split their funds equally between the two currencies in the selected trading pair. They could split at the market rate or use a limit order to split at any rate. To aid this decision, help section contained a simple chart showing a yearly average price, a three-month average price, and a distance from the current market price, as shown in Figure 2. Figure 2 is composed of two charts, Litecoin / Bitcoin price movement using weekly candles and the chart on the right, displaying the number of days a price has remained at each price interval over a year, offering an additional representation of the distribution of rates in the past year.

Figure 1: A backtest of the default settings on the LTC/BTC trading pair.

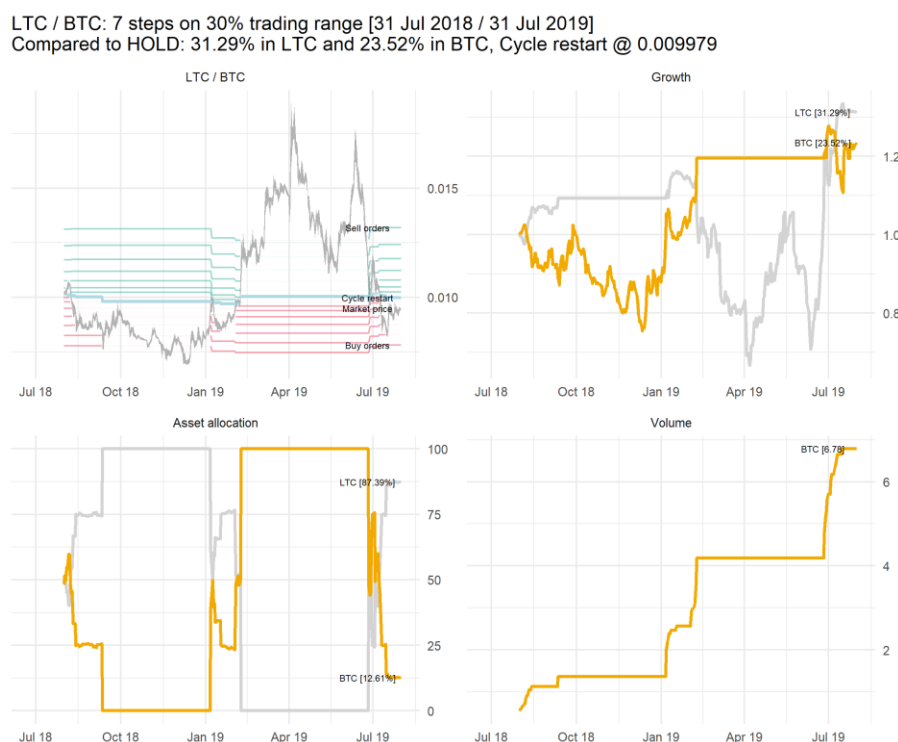
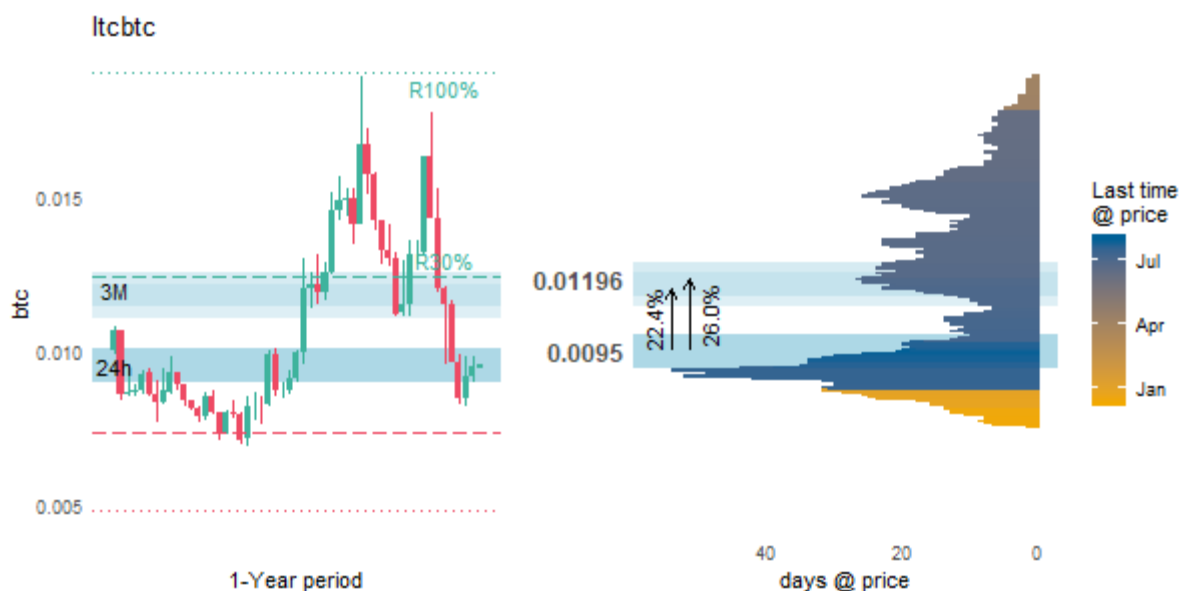


Figure 2: Prices on the Litecoin / Bitcoin market



R 3.4.0 (R Core Team, 2018) was used to conduct the statistical analysis. Package »dplyr« (Wickham et al., 2018) was used for data manipulation, and »ggplot2« (Wickham, 2016) to produce the figures.

3 Results

Descriptive statistics are presented in Table 1. A total number of (sub) accounts used was 191 and a total of 235 active trading pairs because users could change or add a trading pair. As intended most trading pairs were bitcoin denominated trading pairs (194, 86 euro-denominated pairs and 31 dollar-denominated pair), while the average invested amount was 1.64 BTC, 926 EUR, and 1625 USD respectively (see Table 1). Most users selected Ether / Bitcoin trading pair, followed by XRP, Litecoin in Bitcoin Cash for the Bitcoin denominated pairs.

Table 1: Funds allocated by trading pair denomination

	n	mean	sd	min	max
Bitcoin	118	1.64	9.05	0.0114	78.8
Euro	86	926	1017	32.3	6961
US Dollar	31	1625	1544	62.5	5606
Duration	191	142.5	150.6	0	365

Average time spent actively using on-line application was 142 days, 41% of accounts were still active after the test period. Average monthly gain on bitcoin-denominated trading pairs was 4.65% (5.53% Bitcoin Cash, 5.11% Litecoin, 4.64 XRP, and 3.51% when paired with Ether).

4 Conclusion

Currently, cryptocurrencies are held for investment purposes rather than being used for transactions and thus as a medium of exchange. For cryptocurrency to become widely adopted as a means of payment, it first needs to go through a very volatile period because speculative traders see long-run future value in the cryptocurrency. In order to soften transition from speculative asset to medium of payment a trading strategy was proposed, which provides liquidity and reduces volatility. Similar to pairs trading strategy, the proposed solution is based on cointegration. The main difference is that proposed solution does not include shorting, but holding cryptocurrencies, thus increasing the total available cash and adding to the equilibrium price. Results from an ongoing experiment suggest that the proposed trading strategy is appealing for about 40% of cryptocurrency investors, as the struggle against volatility problem is accompanied by significant financial gains. Average monthly gain on bitcoin-denominated trading pairs was 4.65% (5.53% Bitcoin Cash, 5.11% Litecoin, 4.64 XRP, and 3.51% when paired with Ether). As more traders adopt the proposed trading strategy, a transition from cryptocurrencies as speculative assets to medium of payment is well in sight.

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